

Project Design Phase Solution Architecture

Date	15 February 2025
Team ID	6ab221fbe880316a482c6241
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture for this project bridges the business problem (unauthorized users viewing or editing sensitive records) with a ServiceNow-based technical solution. This project implements a script-controlled Access Control List (ACL) that restricts users from viewing records unless they meet a specific condition: only users belonging to the EEE branch can view the records, while all other users are denied access. Its goals are to:

- Find the best tech solution to the business problem: restricting record access so that only users from the EEE branch can view the records.
- Describe the structure, characteristics and behavior of the ACL solution to project stakeholders (admins, faculty, students).
- Define features, development phases and solution requirements: table selection, ACL rule (Table, Operation = read), EEE branch condition, ACL script and testing.
- Provide specifications according to which the solution is defined, managed and delivered: script logic (answer = true/false), testing by impersonating an EEE user and a non-EEE user, and deployment through an Update Set.

Example – Solution Architecture Diagram:

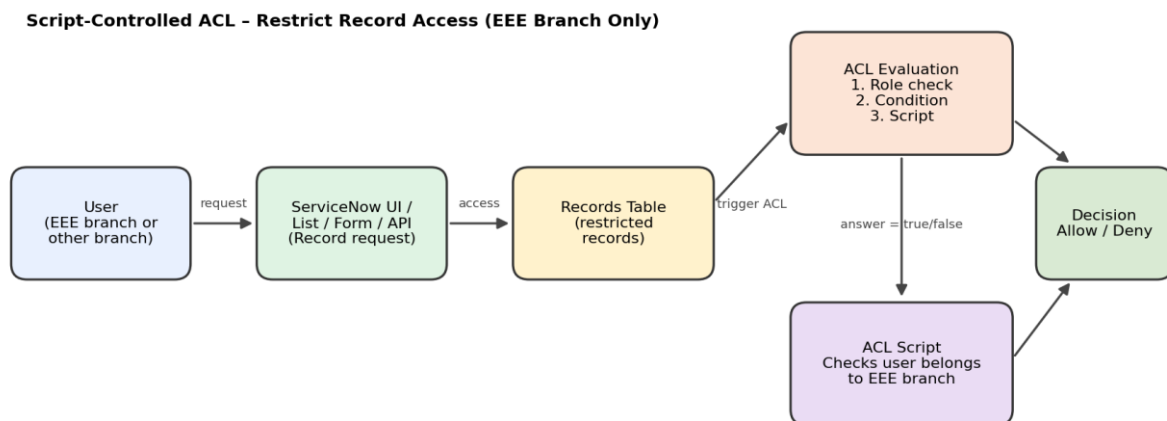


Figure 1: Architecture and data flow of the script-controlled ACL solution

Components and Flow:

- A user opens the records list or form in the ServiceNow platform.
- The platform checks the table-level ACL for the read operation.
- ACL evaluation runs in order: role check, condition, then the advanced script.
- The ACL script checks whether the logged-in user belongs to the EEE branch using `gs.getUser()`.

- The script sets answer = true (allow) or answer = false (deny); records are hidden from users outside the EEE branch.

Sample ACL Script:

```
(function () {  
  // Allow admin users full access  
  if (gs.hasRole('admin')) {  
    return true;  
  }  
  // Allow only EEE branch users to see EEE records  
  if (gs.hasRole('bb1')){  
    return true;  
  }  
  // Deny access for all others  
  return false;  
})();
```